

Machine Learning Applications Using Production Records at Air Force Maintenance Depots

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Abstract:

The Ogden Air Logistics Complex at Hill Air Force Base performs depot maintenance for a wide range of systems. The depot maintains a database to record work, manage logistics, forecast future workload, and facilitate report generation. This study investigates big data and machine learning applications within these data. Findings show various categories of data with sufficient quantity for big data analysis. Weaknesses of the database include infrequent and unreliable connections to legacy systems, a lack of automated user notifications, and low-quality labor hour data. Suggestions for better utilizing the current data include developing data cleaning and imputation tools to improve labor time reporting, implementing user alerts for anomalous data according to a predictive maintenance framework, developing a digital twin scheme to rapidly analyze and forecast base-wide maintenance conditions, and researching the feasibility of generating preliminary workload plans and schedules using reinforcement learning.

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Abstract—The Ogden Air Logistics Complex at Hill Air Force Base performs depot maintenance for a wide range of systems. The depot maintains a database to record work, manage logistics, forecast future workload, and facilitate report generation. This study investigates big data and machine learning applications within these data. Findings show various categories of data with sufficient quantity for big data analysis. Weaknesses of the database include infrequent and unreliable connections to legacy systems, a lack of automated user notifications, and low-quality labor hour data. Suggestions for better utilizing the current data include developing data cleaning and imputation tools to improve labor time reporting, implementing user alerts for anomalous data according to a predictive maintenance framework, developing a digital twin scheme to rapidly analyze and forecast base-wide maintenance conditions, and researching the feasibility of generating preliminary workload plans and schedules using reinforcement learning.

Keywords—aircraft maintenance, machine learning, neural networks, data imputation, predictive maintenance, digital twin, reinforcement learning

I. INTRODUCTION

The Ogden Air Logistics Complex (OO-ALC) at Hill Air Force Base performs depot maintenance on various weapon systems including the F-35, the F-22, the F-16, the A-10, the C-130, the T-38, and the Minuteman III intercontinental ballistic missile (ICBM). The complex is also a leading provider of pneumatics, secondary power systems, composites, and ICBM rocket motors for the Department of Defense (DOD) [1]. Several decades of this work have been digitally recorded in a structured query language (SQL) database called Impresa.

Recent advancements in big data analytics and machine learning can improve maintenance operations at OO-ALC. Potential opportunities focus on saving time and effort in monotonous tasks in order to enable airmen to focus on more challenging workloads. This work is a non-exhaustive survey that describes a preliminary investigation to determine the potential for such opportunities. Complex analysis was not permitted due to data security constraints requiring the data to remain on low compute-capable machines but should be conducted in future work.

This paper is organized as follows: Section II discusses the methods used to understand the database and explore its current capabilities. Section III reports the condition of the dataset and a preliminary feasibility analysis for ML applications. Sections IV-VII describe foundational ML technologies

and corresponding opportunities using these data. Finally, Sections VIII and IX discuss recommended future work and conclusions respectively.

II. METHODS

This section discusses the methods used to understand the database and find new data opportunities therein.

A. User Interviews

A series of stakeholder interviews with subject matter experts (SMEs) served as the primary means for understanding the data. The first interviews began with personal contacts in several groups at OO-ALC and continued with their referrals. The participants in this series of interviews included planners, process engineers, and others with no leadership responsibilities. The second series of interviews focused on the database's steering committee. These interviewees typically had more work experience and leadership responsibilities. In total, approximately 20 SMEs from several major career expertise (engineering, planning, scheduling, analysis, and maintenance) provided a wide range of viewpoints and opinions.

The conversations followed a similar pattern that included the following topics and questions:

- 1) Brief introduction to this research.
- 2) What is your job description and/or responsibilities?
- 3) What is your typical interaction with the data?
- 4) What kind of data do you use?
- 5) (If in person) Can you show me your workflow?
- 6) What pain points do you have with the system? What would you change?
- 7) Here are some ideas previous interviewees have mentioned, what do you think? In this section, describe how ML could apply.
- 8) Now that you've heard other ideas, what else do you think can be done?

B. Data Review

A preliminary review of the data removed all tables that had not been updated in two years and tables with no entries. Due to the importance of large data quantities for ML and the labor required to review all the data, the manual analysis only reviewed the tables with over a million entries. A basic quantitative review constructed histograms of the size and last

update time for each table. Machine learning opportunities using these data likely occur in tables that are regularly updated and consequently contain a large number of rows.

III. DATA CONDITION

Users typically access and modify data using java or web-based applications, which a local contractor maintains. These programs extract data and arrange them into preset spreadsheet-like reports. Reports include purchase orders, supply information, shipping details, work orders, and many other similar types of data.

Preliminary quantitative reviews show early promise for ML tasks. Fig. 1 shows a two-dimensional histogram of the number of entries in the tables and the table update frequency. Furthermore, the data are already configured in SQL tables in either numerical or categorical formats. Thirty percent of the tables are either over 100,000 entries or updated weekly. This quantity and frequency of data show potential benefits for advanced analytics.

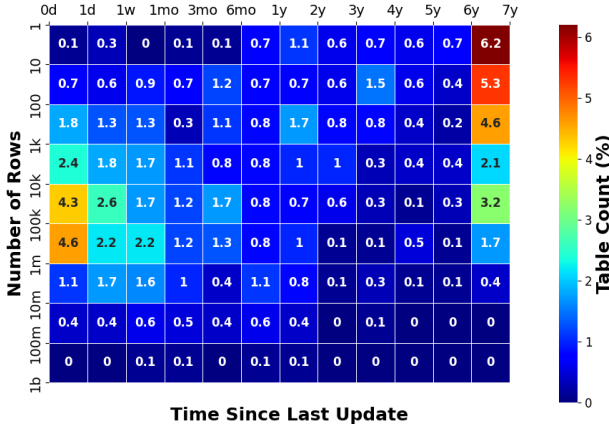


Fig. 1. Heatmap of the data table size and update frequency.

Despite the upfront promise of the data quality and quantity, there are several adverse characteristics of the system. First, low access bandwidth creates strain on production performance. Increased load due to online ML tasking may further inhibit production without architecture considerations while the bandwidth is low. Second, the database is a meta-database that integrates various legacy database systems that update nightly. These nightly updates regularly fail and create inaccurate data.

Finally, many data categories are either duplicated or simply unknown. The interviews revealed the average user may not be aware of subsets of data. Interviewees commented on the difficulty of training new employees due to the disjoint nature of the current reports or the need to pull several different reports to answer relatively simple questions. This data duplication and complexity show a need to better catalog and report the data. Future ML exploration would benefit from a summary of the available data, what reports they generate, and which SQL tables they refer to.

IV. DATA ANOMALY IMPUTATION

Advanced data analysis requires large amounts of high-quality data [6]. It is therefore important to ensure clean data samples and minimize sample rejection. The task of finding unexpected samples in data is called anomaly detection. Anomaly detection techniques range in complexity from simple techniques, like visual inspection, and thresholding, to complex techniques, like clustering and deep classification [4].

When analysts detect anomalous samples, they may consider removing those samples from the dataset. However, such a method discards valuable data and may miss the nuance in what the data represents [14]. Imputation aims to salvage the remaining data by estimating the missing values.

There are three major methods of data imputation [15]:

- 1) Mean/mode: Simply replace values with the mean or mode of the remaining values in the column.
- 2) Classical regression: Replace values using traditional ML and statistical techniques such as linear regression, k-Nearest Neighbor (KNN), or Support Vector Machines (SVM).
- 3) Deep learning: Use deep neural networks, whether discriminative or generative, to predict the missing values.

A. Labor Hour Imputation

Maintenance technicians record the time required to complete a task by scanning when they begin and end working on the task. These reports influence budgeting, product ordering, workload planning, and many other administrative tasks at maintenance depots. Therefore, high-quality labor hour data are necessary. Many interviewees discussed the low-quality nature of these data due to technicians improperly submitting their time. Whether or not this is intentional, the reported data corrupts downstream reporting.

Multiple factors and expected consequences exist for labor time corruption. Unstable connections to the database can set a technician's reported labor time to 0 when sent to the server, which completely loses these values. A second contributor to near 0 scan times is batch scanning, which is when a technician completes several tasks and scans them all at once. Further corruptions may influence technicians to match the submitted time with the expected time for a process. One interview reported technicians could be incentivized to submit their labor hours to improperly receive overtime opportunities.

Fig. 2 shows labor hour efficiency histograms. The x-axis shows the labor efficiency rate and the y-axis shows the number of jobs submitted with that efficiency. Labor hour analysts assume a normal distribution to these data, so the author superimposed manually-fit gaussian curves to show how the real data diverges from an "ideal" data distribution. First, the frequency spike at zero labor efficiency verifies the input data are either not validated at submission or dropped when transferred to the server. Next, there is a noticeable difference in frequency drop-off on either side of the mean. This could be driven by technicians artificially scanning before the expected time for the task. The high value at 0.15 hours worked vs. expected shows possible evidence of batch scanning.

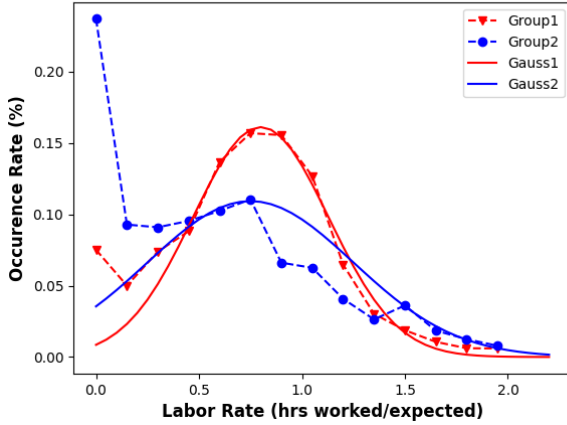


Fig. 2. Labor hour records from all shops for 1 day (red) and one shop for 2 weeks (blue) with aligned gaussian distributions show data discrepancy near zero.

Labor hour reports could benefit from anomaly detection and imputation. This technique may correct poor scanning discipline, estimate corrupted values, and potentially create a more normal distribution of reported labor hours. A promising approach to correcting these data is to train an ML model using verified labor hour submissions, then use the learned model to predict potentially anomalous entries [15]. Inputs to such a model may include production number (PDN), process identification, time of day work began, time of day work finished, plant id, employee experience, and day of the week. Other useful data can be discovered via feature selection algorithms.

V. PREDICTIVE MAINTENANCE

Predictive maintenance (PdM), also known as Condition Based Maintenance (CBM or CBM+), uses sensor data from a part or system to make maintenance decisions. These programs benefit from reducing waste and downtime by avoiding potentially unnecessary routine maintenance while reducing failures in the field. This type of maintenance is suitable for applications where available data can cost-effectively predict failures.

Commercial airlines claim “predictive maintenance prompts are valid more than 95 percent of the time ... [and estimate] predictive maintenance reduced operation disruptions by 30-50 percent” [18]. The DOD has directed the services to use PdM from as early as 2002 and instructed PdM to be codified in 2007 [8], [18]. The Air Force has made concerted efforts in recent years to expand PdM to all aircraft [9]. Preliminary results of PdM of 8 C-5 cargo aircraft and 10 B-1 bombers show an estimated cost avoidance of \$6.2 million [18].

There are five capability levels of PdM [13]:

- 0) No PdM: Maintenance is only run on a regular schedule.
- 1) Visual inspection: Alerts occur when an inspector detects an issue based on their expertise.

- 2) Instrument inspection: Alerts occur after periodic inspection and instrumented testing.
- 3) Continuous condition monitoring: Alerts occur at any time during operation based on simple rules.
- 4) Continuous analytics: Alerts occur at any time during operation using more advanced regression techniques.

Fig. 3 shows these levels represented graphically. Monitoring frequency (x-axis) represents how often the system is analyzed, where continuous monitoring during operation is ideal. Monitoring complexity (y-axis) is the capability of the monitoring systems to predict future failure.

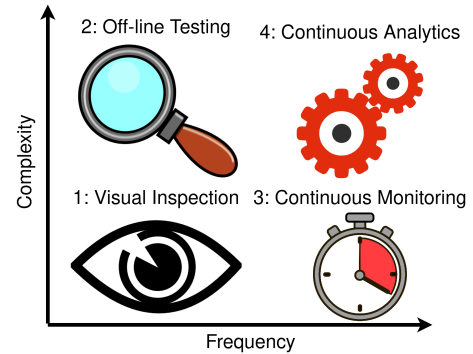


Fig. 3. Graphical representation of PdM levels.

DOD instructions [8] and DOD Inspector General (DODIG) audits [9] outline six functional activities in PdM and CBM+ (Fig. 4). This author suggests a missing functional activity before analysis occurs. Many stored data are not in a usable format for advanced analytics [2], [7]. Preprocessing the data may require removing outliers, filling in missing information, standardizing inputs, or labeling samples [21]. Therefore, an additional step to preprocess the data must occur before conducting analysis. This report focuses on the potential analysis of existing stored data (Activity 4).

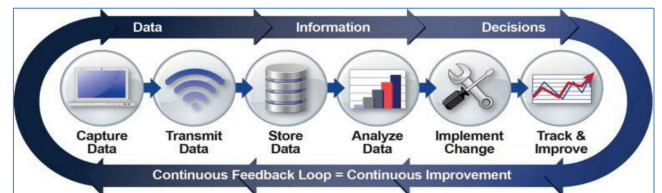


Fig. 4. Six Functional Activities of CBM+ [8].

A. Pricing Anomaly Notification

The database tracks thousands of parts and processes. Manually tracking these items individually is both inefficient and ineffective. Several interviewees remarked it is easy to identify when samples are anomalous, but it is a “needle in the haystack” process of finding them. The interviewees were unaware of automated notification tools that could generate alerts of anomalous data.

Tracking pricing data is an activity that would benefit from such notification. A planner recently discovered a part that

unknowingly increased in price by 1,300 percent in two years. Current methods of finding these price increases are manual and unnecessarily time-consuming. They could be described as PdM Level 1. Continuous condition monitoring, PdM Level 3, can track the price of each part against a simple inflationary model and alert users when anomalies occur (Fig. 5).

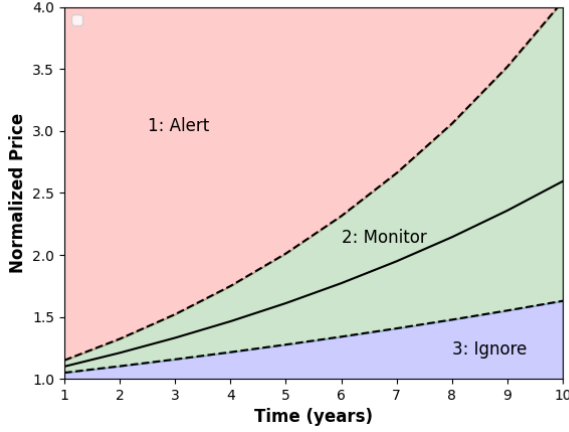


Fig. 5. Inflationary forecasting can alert purchasers of unexpected price increases. Supplies in regions 2 and 3 (green and blue) are expected and would be ignored. Alerts can notify when a supplier's cost reaches region 1 (red).

B. Labor Hour Divergence Notification

A second low-complexity data analysis tool would aid in setting the expected labor hours for a task. This task assumes that reported labor hours are accurate and follow an understood probability distribution function. Each year, analysts manually review 20 percent of maintenance processes and update the expected labor time of each process. Processes cycle through this review sequentially, thus assuring each process is reviewed every five years. This review is comparable to PdM Level 2 and may benefit from PdM Levels 3 or 4 monitoring to flag the most out-of-distribution processes for immediate review. There are several tests within classical statistics that can verify if recent data matches the expected distribution. Automated notification can alert analysts when a process has significantly deviated from its established parameters.

VI. DIGITAL TWINS

Digital Twins (DT) are simulated representations of real-world entities using live sensor data. The term was first coined by NASA by saying, "A Digital Twin is an integrated multiphysics, multiscale, probabilistic simulation of an as-built vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding flying twin" [11]. The application of DTs uses physical data to inform simulation models to consequently drive decisions and actions for the physical entity. Fig. 6 shows a digital twin example with the space shuttle.

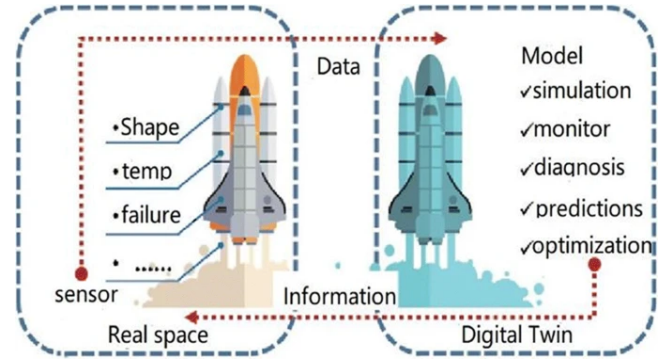


Fig. 6. Digital twin information flow paradigm [12].

Research indicates that relevant benefits include live monitoring, scenario risk assessment, better inter-team synergy and collaborations, more efficient decision-making, and better communication [20]. These benefits stem from all the data for a given system, both live and historical, being organized in one central location. Models of the data then enable operational status visualization, expected projections, and what-if analysis.

The Under Secretary of Defense for Research and Engineering released the "Digital Engineering Strategy" in 2018 that promotes DT technology. Among other goals, this strategy calls for "[Making] use of data to improve awareness, insights, and decision making". Properly utilizing the data will enable "DOD's vision to build an enterprise capability that securely leverages data and analytics to enable insights and achieve faster and better data-driven decisions" [19]. The expected result of enterprise-level data visualization is an improved ability to assess current status and optimize operations in shorter periods of time.

A. Digital Twinning Maintenance Depots

Rapid visualization for status and reporting are weaknesses of the database's current functionality. Generating cohesive reports typically involves combining several web application reports and SQL data pulls. These data then must be post-processed manually in a time-consuming and error-prone process that requires expert knowledge.

Constructing a digital twin of the depot can alleviate these problems. The multi-faceted benefits of this approach include immediate data visualization, reduced experience requirements due to data centralization, and scalable model forecasting. Leadership, analysts, and planners could view all the available data pertaining to an on-base entity at a glance. Fig. 7 shows an example visualization of this concept.

As Fig. 7 shows, the user is viewing the current status of the main repair hangar. In this hangar, the user sees the aircraft docks and can immediately determine the repair status of the occupying aircraft. Color coding indicates repair time compared to the planned time. Further examining the aircraft could inform the user of the procedures and materials that remain until the repair procedures conclude (Fig. 8).

A final feature digital twinning would enable is rapid what-if analysis. Building the digital twin requires centralizing all

View Mode: Building Status		Entity Select: Main Hangar	
Sub Entities			
1:  HF-117 Finished: 4/15/23	2:  Empty Expected: 3/22/23	3:  AZ-867 Finished: 3/17/23	4:  WA-5309 Finished: 7/4/23
Machines Supplies Personnel Repair Docks			
1: HF-117 2: Empty 3: AZ-867 4: WA-5309 ...			
20:  HF-123 Finished: 10/4/23	21:  HF-345 Finished: 8/5/23	22:  HF-456 Finished: 12/10/23	23:  HF-789 Finished: 6/21/23

Fig. 7. A DT of repair hangars could show maintenance repair progress and forecasting of all aircraft within a specific building. Color coding can indicate progress compared to expectations. Further details at the building level could include budget, supplies, personnel, and process capabilities.

View Mode: Aircraft Status		Entity Select: HF-117	
			
Tail Number: HF-117	Home Unit: Hill AFB	Process: Depot Maint. 1	
Date In: 2/3/23	Date Due: 5/1/23	Date Exp: 4/15/23	
Depot Maintenance 1 Overview			
Financial Information			
Supplies Status			
What-If Analysis			
Schedule			
Current Process: Process 1	Current PDN: 123456	Technician: L.Jenkins	
Process Begin: 3/4/23	PDN Begin: 3/9/23 9:42	Location: Main Hangar-1	
Process End: 4/6/23	Labor Time: 5/32 hrs	Supplies	
Inspect	Process 1	Process 2	Review
Feb. 23	Mar. 23	Apr. 23	May 23

Fig. 8. Detailed views of a specific aircraft can readily show current schedules, financial information, production status, etc.

the data for a particular entity and forming a representative model of that entity. With the model in place, analysts can modify parameters and immediately see the impact of future maintenance operations. Configurable parameters can include part pricing and supply, repair time mean and standard deviation, available repair docks, etc. Forecasting the analysis enables rapid response to unexpected operations impediments.

Enterprise-level implementation of such a technology would be an immense endeavor requiring a multi-year, multi-million dollar effort. The benefits of this concept are further hampered by deficient bandwidth from the data servers. A potential development path to ease the time to benefit and server strain is the minimally viable product (MVP) approach. In this scheme, preliminary designs of low functionality begin with subsets of parts and features or low complexity analysis. Features are added in iterations according to customer demand.

VII. REINFORCEMENT LEARNING

Reinforcement Learning (RL) is an ML paradigm inspired by behavioral psychology where an organism learns through trial and error [22]. Using this paradigm, agents receive a state

from the environment, choose how to act on that state, and receive a reward based on the outcome of the selected actions [5]. Typical RL applications include robotics [17], control systems [16], and financial markets [10].

Recent work in RL shows promising applications in workload planning and scheduling in the Air Force. The DAF-MIT Artificial Intelligence Accelerator produced the scheduling tool Puckboard.AI that creates AI-powered training schedules for student pilots [23]. Another example is a recent data challenge hosted by the Air Force Research Lab, titled the Airlift Challenge, that explored applying RL to cargo delivery operations under uncertain conditions (Fig. 9) [3].

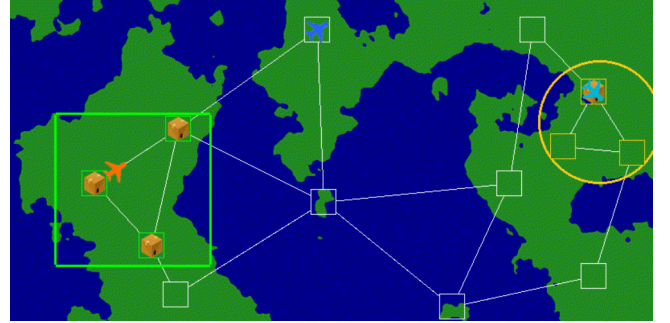


Fig. 9. The Airlift Challenge requires agents to transport packages from the green region to the yellow region while the paths (white) experience intermittent drop-out.

A. Planning and Scheduling using RL

Given this recent success in both industry and defense, RL may be beneficial for maintenance depot workload planning. Such an application could develop preliminary workload plans that would then be fine-tuned and approved by professional planners. This work would be near state-of-the-art and would require extensive research and development. The potential time-saving and robustness benefits merit consideration despite the expected upfront costs.

VIII. FUTURE WORK

The size and importance of these data present many opportunities for improved data analytics. The first step for validating this work is to demonstrate a low-risk, short-term proof of concept. Labor hour anomaly detection and correction, as discussed in Section IV-A, is a strong candidate for this effort due to the large quantity of available data, wide-spread benefit, and potentially low-complexity implementation.

Next, a severe limitation of this work was the lack of overlapping knowledge in AI/ML and depot maintenance operations between the researcher and the interviewees. Future investigations of these data will benefit from exposing the data scientists to maintenance operations and conducting an ML crash course for SMEs. Building this knowledge overlap will generate higher-quality ideas from both parties.

A final and ambitious suggestion for future work is holding a hackathon with the data. Hackathon competitions bring SMEs and programmers together to intensively develop new

capabilities in a short period of time. Researched benefits of hackathons include cultural change, talent identification, technology innovation, and platform assessment [24]. A hackathon of the data will likely yield innovation but will experience significant security and logistical hurdles.

IX. CONCLUSION

Maintenance data at Air Force logistics complexes are underutilized and ready for machine learning innovation. Legacy systems and rigid organization inhibit the data's usefulness and burden users with unnecessary monotony. OO-ALC has a significant need to collect more meaningful data, repair current data, and automate data processing to better analyze depot operations. Leveraging modern analytics on these data is a prime candidate for a significant force multiplier.

This study comprises an analysis of the data for ML opportunities from SME interviews. Four technologies of differing complexity and impact show a breadth of potential development. The author recommends immediately pursuing labor hour imputation and automated anomaly notification. Efforts to discover additional innovative ideas will benefit from an action group of cross-trained ML and Logistics SMEs. Finally, OO-ALC should consider prototyping the digital twin dashboard at a hackathon event.

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